Matrices

Read the shape. Understand the action.

Basic Mathematics ▪ Lecture 1

$$\begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}$$

The story of the lecture

Read the object → Operate entrywise → Compose actions → Notice structure → Transform the problem

A matrix is useful because it keeps structure visible while we calculate.

Three 45-minute arcs

Arc I — Language

shape, entries, rows, columns

addition and scaling

Arc II — Action

row $\times$ column

composition and order

Arc III — Structure

identity, transpose, diagonal

row operations

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What matrices are

From a rectangular array to a mathematical object

A matrix can play several roles

$$A = \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}$$

data **coefficients** **relations**

transformation **geometric action**

The entries alone do not tell the whole story; the interpretation comes from the problem.

The first skill is reading

What is the shape? → What do entries mean? → How should rows and columns be read?

Calculation comes after orientation.

General notation

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Shape

$$A \in \mathbb{R}^{m \times n}$$

m rows

n columns

Address

$$a_{ij}$$

row i, column j

Read one matrix carefully

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}$$

Shape

$M \in \mathbb{R}^{2\times 3}$

Selected entries

$m_{11} = 4$
$m_{12} = -2$
$m_{23} = 5$

Blue: first row. Green: entry in row 2, column 3.

Rows and columns are different objects

Row matrix

$$r = \begin{pmatrix} 4 & -2 & 0 \end{pmatrix}$$

$$1 \times 3$$

Column matrix

$$c = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$$

$$3 \times 1$$

Same numbers, different shape, different algebraic role.

A small gallery of shapes

row

$$\begin{pmatrix} a & b & c \end{pmatrix}$$

1×3

column

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

3×1

square

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

2×2

rectangular

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$$

2×3

Shape determines which operations are possible.

Checkpoint: before touching the entries

1. name the shape;
2. locate entries by indices;
3. separate rows from columns;
4. identify the intended interpretation.

Do not start with

"Which formula should I use?"

Start with

"What object am I holding?"

2 Basic matrix operations

Operations that change entries but preserve shape

What remains unchanged?

$$A \in \mathbb{R}^{m\times n} \xrightarrow{\text{same shape}} A+B,\ A-B,\ \lambda A$$

These operations modify values while keeping every position in place.

Compatibility is not a technicality

Defined

$$(2 \times 3) + (2 \times 3)$$

Every entry has a partner.

Not defined

$$(2 \times 3) + (3 \times 2)$$

Positions no longer match.

Addition is position-by-position matching.

The local rule

Addition

$$A + B = (a_{ij} + b_{ij})$$

Subtraction

$$A - B = (a_{ij} - b_{ij})$$

Scalar multiplication

$$\lambda A = (\lambda a_{ij})$$

one number acts on every entry

Worked example: addition

$$\underbrace{\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}}_{A} + \underbrace{\begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}}_{B} = \underbrace{\begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix}}_{A+B}$$

A		B		$A + B$
$\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$	$+$	$\begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}$	$=$	$\begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix}$

$$2 + 5 = 7 \qquad -1 + 4 = 3 \qquad 0 + (-2) = -2 \qquad 3 + 1 = 4$$

Worked example: subtraction and scaling

$A - B$

$$\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -3 & -5 \\ 2 & 2 \end{pmatrix}$$

$-3A$

$$-3\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}$$

Linear combinations: build a new matrix

$$\alpha A + \beta B$$

$$2\begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -4 & 4 \end{pmatrix}$$

Interpretation

scale two matrices, then combine matching positions

Why familiar algebraic laws survive

$$A + B = B + A$$

$$(A + B) + C = A + (B + C)$$

$$\lambda(A + B) = \lambda A + \lambda B$$

Reason

Each statement is true entry by entry because ordinary numbers satisfy the same laws.

Before matrix multiplication

Entrywise model

This model belongs to addition and scalar multiplication.

Row–column model

Matrix multiplication makes a row interact with a column.

The next operation changes the logic, not only the formula.

Checkpoint: entrywise operations

- same size is required;
- shape is preserved;
- every entry keeps its address;
- familiar arithmetic laws pass through.

Position is the organizing principle.

3 Matrix multiplication

Rows meet columns, and order becomes meaningful

Why multiplication needs a new rule

Addition

Matching positions interact.

Multiplication

One row interacts with one column.

A product combines information across positions.

Dimension rule: inner match, outer remain

$$(m \times n)(n \times p) = m \times p$$

Inner dimensions

Same length: one row of A meets one column of B.

Outer dimensions

Number of rows and columns in the product.

One output entry

$$\underbrace{\begin{pmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{pmatrix}}_{\text{row } i \text{ of } A} \underbrace{\begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix}}_{\text{column } j \text{ of } B} = c_{ij}$$

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

The whole product is a grid of local interactions

$$C = AB = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{pmatrix}$$

Each location

c_{ij} records row i of A interacting with column j of B.

Global from local

The product is assembled from these row–column answers.

Worked example: set up the product

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}$$

Shape check

$$(2 \times 2)(2 \times 2) = 2 \times 2$$

Plan

Four output entries require four row–column products.

Worked example: the first row

c_{11}

$$\begin{pmatrix} 1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = 1 \cdot 2 + 3 \cdot 5 = 17$$

c_{12}

$$\begin{pmatrix} 1 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = 1 \cdot 0 + 3(-1) = -3$$

The same row produces a full row of the product.

Worked example: the second row and the result

$$c_{21} = (-2) \cdot 2 + 4 \cdot 5 = 16$$

$$c_{22} = (-2) \cdot 0 + 4(-1) = -4$$

Assembled product

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}$$

Reverse the order, ask a different question

AB

$$\begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}$$

BA

$$\begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}$$

$AB \neq BA$ **in general.**

Why order matters conceptually

$$\boldsymbol{x} \xrightarrow{B} B\boldsymbol{x} \xrightarrow{A} A(B\boldsymbol{x})$$

$$\boldsymbol{x} \xrightarrow{A} A\boldsymbol{x} \xrightarrow{B} B(A\boldsymbol{x})$$

Changing the order changes the process being performed.

Sometimes only one order exists

Defined

$$A \in \mathbb{R}^{2\times 3}, \quad B \in \mathbb{R}^{3\times 1}$$

$$AB \in \mathbb{R}^{2\times 1}$$

Not defined

$$(3 \times 1)(2 \times 3)$$

The inner dimensions 1 and 2 do not match.

Matrix times vector

$$A \in \mathbb{R}^{m\times n}, \qquad \boldsymbol{x} \in \mathbb{R}^n$$

$$A\boldsymbol{x} \in \mathbb{R}^m$$

Interpretation

The matrix receives an input vector and produces an output vector.

Shape logic

Input length n must match the number of matrix columns.

Columns explain the action

$$A = \begin{pmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_n \end{pmatrix}, \qquad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n$$

The output is a linear combination of the columns of A.

Associativity and order

Associativity

$$(AB)C = A(BC)$$

Grouping may change.

Order

$$AB \neq BA \quad \text{in general}$$

Order remains fixed.

Checkpoint: multiplication

1. check inner dimensions;
2. predict the output shape;
3. compute row by column;
4. keep the order fixed;
5. interpret the product as composition.

Each entry is local; the transformation is global.

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Special matrices and visible structure

Patterns are mathematical information, not decoration

Structure changes the calculation

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Visible immediately

no off-diagonal interaction

Consequence

powers, products and interpretation become transparent

Seeing the pattern is part of solving the problem.

Zero and identity: two neutral objects

Zero matrix

$$A + 0 = A$$

neutral for addition

Identity matrix

$$I_m A = A, \qquad A I_n = A$$

neutral for multiplication

Identity sizes must fit

$$A \in \mathbb{R}^{m\times n}$$

Left identity

$$I_m A = A$$

I_m has shape $m \times m$

Right identity

$$AI_n = A$$

I_n has shape $n \times n$

The identity is conceptually one object, but dimensionally many matrices.

Transpose: rows become columns

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \qquad \longrightarrow \qquad A^{\mathsf{T}} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

$$2 \times 3 \qquad\qquad\qquad 3 \times 2$$

Transpose rules and symmetry

$$(A^\mathsf{T})^\mathsf{T} = A$$
$$(A+B)^\mathsf{T} = A^\mathsf{T} + B^\mathsf{T}$$
$$(AB)^\mathsf{T} = B^\mathsf{T} A^\mathsf{T}$$

Symmetric matrix

$$A^\mathsf{T} = A$$

entries mirror across the main diagonal

Diagonal matrices

$$D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$$

Visible pattern

Only the main diagonal may contain non-zero entries.

Repeated products of a diagonal matrix

$$D = \operatorname{diag}(3, -1, 2)$$

Second product

$$D^2 = \operatorname{diag}(9, 1, 4)$$

Third product

$$D^3 = \operatorname{diag}(27, -1, 8)$$

Each diagonal position evolves separately.

Patterns worth noticing

Zeros

sparse, triangular, diagonal

Repetition

equal rows, columns, blocks

Dependence

proportional rows or columns

Symmetry

mirrored entries

Use the pattern

State what becomes simpler and why.

Checkpoint: see before you compute

look for patterns → explain their meaning → use them to simplify

Noticing structure is mathematical understanding.

5 Row operations and problem solving

Transform the representation to reveal hidden relations

Why change the matrix at all?

Not the goal

Make the matrix look different.

The real goal

Reveal dependence, simplify structure, prepare a system for solving.

A row operation is a controlled change with a purpose.

The three elementary row operations

Swap

$$R_i \leftrightarrow R_j$$

reorder equations

Scale

$$R_i \leftarrow \lambda R_i, \quad \lambda \neq 0$$

change normalization

Combine

$$R_i \leftarrow R_i + \lambda R_j$$

remove repeated information

A row operation should tell a story

matrix before → chosen operation → row arithmetic → matrix after

The interpretation is the final step: what did the transformation reveal?

Row-operation example: before and intention

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}$$

Observation

The first two entries of R_2 are exactly twice the first two entries of R_1.

Intention

Remove that repeated part and see what remains.

$$R_2 \leftarrow R_2 - 2R_1$$

Row-operation example: arithmetic made visible

$$\underbrace{(2,4,3)}_{R_2} - 2\underbrace{(1,2,-1)}_{R_1} = (2-2,\ 4-4,\ 3+2) = \underbrace{(0,0,5)}_{R_2^{\mathsf{new}}}$$

Cancelled

The repeated contribution in columns 1 and 2.

Exposed

A remaining contribution in column 3.

Row-operation example: before, operation, after

before

$$\begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$

$$\Longrightarrow$$

after

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}$$

The new row shows exactly what was not already explained by the first row.

The three layers of the same step

1. Arithmetic	2. Transformation	3. Interpretation
subtract two row vectors	replace R_2 by a new row	reveal dependence and residual information

A good solution makes all three layers visible.

Column vectors are matrices

Column form

$$\boldsymbol{u} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} \in \mathbb{R}^{3\times 1}$$

Row form

$$\boldsymbol{u}^{\mathsf{T}} = \begin{pmatrix} 2 & -1 & 4 \end{pmatrix} \in \mathbb{R}^{1\times 3}$$

Transpose changes the algebraic role of the same coordinates.

Lecture map: one connected story

read shape → operate entrywise → compose actions → notice structure → simplify rows

The calculation matters; the interpretation connects the calculation to the problem.

Read the shape.

Then calculate. Then interpret.